Aryan Naik

aryannaik1205@gmail.com, github.com/RobotsAndAryan

EDUCATION

MEng Robotics and Artificial Intelligence, University College London (UCL), 2025 to 2029 (expected)

SKILLS

Extended Reality and Simulation: OpenXR, Unity C#, NVIDIA Isaac Sim, digital twin simulation, immersive simulation, Meta Quest 3

Robotics and Software: ROS 2 Jazzy, FastDDS middleware, distributed autonomy, spatial computing

Electronics and Hardware: breadboard circuit wiring, microcontroller logic, hardware integration

Design: CAD, Autodesk Fusion

PROJECTS

Aegis Swarm (github.com/RobotsAndAryan)

- Accomplished distributed drone swarm autonomy as measured by successful coordination across multiple agents by architecting a ROS 2 Jazzy system with FastDDS middleware configuration
- Accomplished high-fidelity digital twin simulation of the swarm environment as measured by full pre-deployment virtual replication of swarm behaviour by building a NVIDIA Isaac Sim simulation environment, supporting industrial workflow validation without physical hardware risk
- Accomplished immersive mixed reality teleoperation of the swarm as measured by real-time command transmission to all active swarm agents simultaneously by developing a Meta Quest 3 OpenXR interface in Unity C#, applying enterprise spatial computing techniques relevant to workforce training and operational deployment

Robotics Challenge (github.com/RobotsAndAryan)

- Accomplished functional electronics hardware integration as measured by a verified working circuit under microcontroller logic execution by completing breadboard wiring and embedded control programming
- Accomplished physical component design as measured by parts produced via CAD modelling in Autodesk Fusion by designing mechanical assemblies for the robotics platform

ADDITIONAL INFORMATION

Coursework and project work spans hardware integration, immersive simulation, and digital twin development with direct relevance to industrial and enterprise deployment contexts including aerospace, defence, and manufacturing sectors. Practical experience building extended reality interfaces and simulation environments aligns with XR Industries' focus on workforce training applications and reducing operational risk through virtual pre-deployment testing.

Aryan Naik aryannaik1205@gmail.com github.com/RobotsAndAryan

Hiring Team XR Industries

Dear Hiring Team,

I am writing to apply for the Engineer Intern position at XR Industries for this coming summer. Your work developing proprietary simulation environments and digital twin solutions for clients in manufacturing, aerospace, and defence represents precisely the intersection of hardware integration and immersive spatial computing that I find most compelling. The fact that XR Industries builds operational tools that directly modernise industrial workflows, rather than producing technology for its own sake, is what distinguishes you from other firms in this space and makes this placement particularly attractive to me.

My most relevant project is Aegis Swarm, a distributed drone swarm autonomy system I developed using ROS 2 Jazzy with FastDDS middleware for inter-agent communication. I built a digital twin simulation environment in NVIDIA Isaac Sim to validate swarm behaviour before physical deployment, and implemented a mixed reality teleoperation interface on Meta Quest 3 using OpenXR and Unity C#, enabling an operator to monitor and direct multiple autonomous agents through an immersive spatial interface. This project gave me direct, practical experience with the same combination of simulation, hardware integration, and extended reality tooling that sits at the core of XR Industries' product offering.

I would contribute a working knowledge of ROS 2, real-time simulation, and XR development pipelines, alongside hands-on electronics and CAD skills developed through hardware integration work in a competitive robotics environment. I am eager to deepen my understanding of how enterprise-grade XR systems are engineered for reliability and scalability in demanding industrial contexts, and to learn how a professional team approaches the full development cycle from client requirement through to deployment.

Yours sincerely, Aryan Naik